Relation-Aware Multiprototype Learning for Semi-Supervised Hyperspectral Image Classification

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***Abstract*—Pseudo-label learning-based methods have achieved excellent performance by leveraging spatial–spectral features of massive unlabeled samples in hyperspectral image (HSI) classification. Existing methods primarily rely on confidence scores predicted by classification models to produce pseudo-labels. However, both limited labeled samples and complex spatial–spectral information result in these models being prone to overfitting, reducing the reliability of pseudo-labels. In this article, a relation-aware multiprototype learning (RAMPL) method is proposed. First, to extract discriminative spatial–spectral features, a hybrid transformer–CNN module (TCM) is developed, which extracts both global and local features using the attention mechanism and a mixed-scale convolutional block (MSCB). Then, to generate high-confidence pseudo-labels, a multiprototype learning module (MPLM) is proposed to construct multiple class prototypes for each category by exploiting both unlabeled samples with pseudo-labels and a small number of labeled samples, which effectively models the intraclass diversity. Finally, to further enhance feature discrimination, a relation-aware contrastive learning module (RACLM) is proposed to learn the complex spatial distribution difference and spectral variability of the HSI by constructing a relation-aware pseudo-graph. Extensive experiments demonstrate that the proposed RAMPL achieves superior classification accuracy, achieving 85.76%, 84.36%, 93.14%, and 96.13% on the Indian Pines (IP), Houston2013 (H2013), Salinas (SA), and Kennedy Space Center (KSC) datasets with five labeled samples in each category, and surpasses 11 state-of-the-art HSI classification methods.**

***Index Terms*—Class prototype, hyperspectral image (HSI) classification, pseudo-label, semi-supervised learning, transformer.**

I. Introduction

REMOTE sensing hyperspectral images (HSIs) record the complex spatial position distributions and spectral response signals of observed scenes, which can be used to analyze the land-cover category of these observed scenes [1], [2], [3], [4]. With the rapid development of sensor technology and image processing capabilities, HSIs have received widespread attention and found extensive applications in environmental monitoring, mineral identification, and natural disaster prevention [5], [6], [7].

HSI classification strives to identify the land-cover category of pixels by analyzing spatial and spectral information [8], [9], [10], [11], [12]. Due to powerful feature learning and representation capabilities, various deep learning-based technologies have been successfully applied to HSI classification. Supervised learning-based methods generally design various neural networks as feature extractors to extract spatial and spectral information from numerous labeled samples, and achieve satisfactory classification performance [13], [14], [15], [16]. However, obtaining sufficient labeled samples is resource demanding, which leads to these methods failing to extract discriminative features when labeled samples are insufficient, thereby limiting classification accuracy.

Semi-supervised learning-based methods aim to improve the performance and generalization by leveraging spatial and spectral information of unlabeled samples [17], [18], [19], [20], [21], [22], [23]. According to the utilization manner of unlabeled samples, existing methods can be categorized into graph learning-based methods [19], contrastive learning-based methods [20], reconstruction learning-based methods [22], and pseudo-label learning-based methods [17], [18], [24]. Graph learning-based methods typically employ the simple linear iterative clustering algorithm to split the HSI into a series of irregular regions, and these irregular regions are treated as graph nodes for feature extraction [19]. Contrastive learning-based methods construct positive and negative sample pairs from labeled and unlabeled samples. Then, these methods push samples belonging to positive sample pairs closer and constrain those from negative sample pairs far from each other to extract features [20]. Reconstruction learning-based methods often design autoencoder networks to learn spatial–spectral information in a reconstruction manner, followed by finetuning the model's parameters using labeled samples [22].

Pseudo-label learning-based methods preserve unlabeled samples with confidence scores larger than a preset threshold and regard their predictions corresponding to the maximum confidence scores as pseudo-labels. These unlabeled samples and labeled samples are used to optimize the model's parameters in a supervised manner [17]. However, due to complex and diverse land-cover categories, adjacent spatial position information of pixels from the same category exhibits differences, while pixels from different categories have similarities, referred to as spatial distribution difference in this article. In addition, affected by imaging conditions such as

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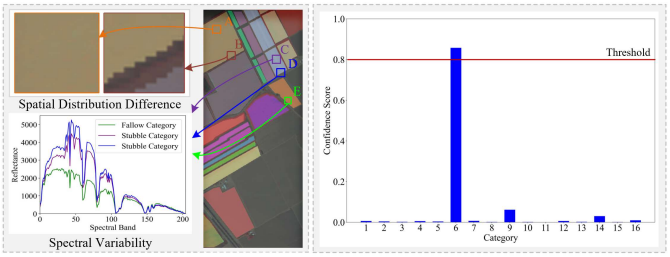
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(a) Complex Spatial and Spectral Information (b) Confidence Score of an Unlabeled Sample

Fig. 1. (a) Hyperspectral pixels from the same category exhibit differences in spatial position distribution and spectral bands, and those from different categories show similarities. (b) PUSL [17] produces the overprediction for an unlabeled sample in experiments with five labeled samples per category on the SA dataset. The ground truth of this sample is category 9, but the prediction is category 6.

temperature and humidity, pixels from the same category exhibit differences in global and local spectral bands, while those from different categories have similar spectral bands, named as spectral variability [25], [26], [27]. As shown in Fig. 1(a), we observe that pixels A and B are from the same category, but their adjacent spatial position information has significant differences. The spectral bands of pixels C and D from the "Stubble" category are distinctly different in local and global perspectives, and the spectral bands of pixel E from the "Fallow" category are similar to those of pixels C and D to a certain extent. Both limited labeled samples and complex spatial–spectral information result in these methods being prone to overfitting, which reduces the reliability of pseudo-labels. In methods that rely on confidence scores to generate pseudo-labels, the generated pseudo-labels may contain incorrect pseudo-labels that do not match the ground truth, illustrated in Fig. 1(b). Leveraging unlabeled samples containing incorrect pseudo-labels to optimize the model's parameters will confuse feature discrimination and degrade classification accuracy.

To accurately identify the land-cover category of hyperspectral pixels, a relation-aware multiprototype learning (RAMPL) method is proposed to improve the reliability of generated pseudo-labels. First, existing transformer-based methods primarily extract global features of the HSI, while failing to effectively capture local spatial and spectral information. To extract abundant spatial–spectral features, we build a hybrid transformer–CNN module (TCM) that incorporates the attention mechanism and a mixed-scale convolutional block (MSCB) to extract both global and local features. Then, to generate high-quality pseudo-labels, we design a multiprototype learning module (MPLM), which leverages unlabeled samples with pseudo-labels and a small number of labeled samples to construct multiple class prototypes for each category. Finally, to further enhance spatial–spectral feature discrimination, we propose a relation-aware contrastive learning module (RACLM), in which a relation-aware pseudo-graph is constructed to model the complex spatial distribution difference and spectral variability of the HSI. The main contributions of this article are summarized as follows.

1) To accurately generate high-confidence pseudo-labels, we propose an MPLM to model intraclass diversity by constructing multiple class prototypes for each category using unlabeled samples with pseudo-labels and a few labeled samples.
2) To further improve the reliability of generated pseudo-labels, we design an RACLM to learn the spatial distribution difference and spectral variability by building a relation-aware pseudo-graph.
3) Extensive experiments demonstrate that RAMPL achieves outstanding classification accuracy and surpasses 11 state-of-the-art methods.

The rest of this article is organized as follows. Section II offers a comprehensive overview of related works. Section III presents the detailed description of the proposed RAMPL. Section IV outlines the experimental settings and discusses the corresponding results. Finally, Section V concludes this article.

II. Related Works

A. Semi-Supervised HSI Classification

Compared with supervised learning-based methods, semi-supervised learning-based methods aim to leverage potential knowledge of unlabeled samples and achieve satisfactory classification accuracy in HSI classification [17], [28], [29]. According to the utilization manner of unlabeled samples, existing semi-supervised learning-based HSI classification methods can be roughly categorized into four groups: graph learning-based methods, contrastive learning-based methods, reconstruction learning-based methods, and pseudo-label learning-based methods.

Graph learning-based methods can be categorized into pixel-based methods and superpixel-based methods based on the generated manner of graph nodes [9], [21], [30], [31], [32], [33], [34], [35], [36]. Pixel-based methods treat each pixel as a graph node, and then use graph neural networks (GNNs) to extract features. To reduce computational resources by decreasing the number of graph nodes, several works propose to split the HSI into a series of irregular regions (known as superpixels) through the simple linear iterative clustering algorithm and consider these superpixels as graph nodes. Wang et al. [9] designed a hypergraph convolutional network (HGCN) with multiple hyperedge fusion, which leverages HGCNs and convolutional neural networks (CNNs) to extract diverse features. Lu et al. [21] proposed a superpixel-guided label propagation module to generate high-quality pseudo-labels. Zhou et al. [30] proposed a CNN-based parallel structure and multihop graphs to capture pixel-level and superpixel-level features, and built a cross-attention fusion module to enhance the fusion features. Qin et al. [31] treated several neighboring pixels as a graph node and then used GNNs to extract features. Xi et al. [32] constructed a cross-scale graph structure to represent spatial–spectral information of the HSI. Yang et al. [33] combined GNNs and CNNs to extract global and local information and proposed a pixel-level feature update strategy to learn pixel representation. Jia et al. [34] proposed a graph-in-graph structure consisting of internal and external graphs to extract local and global features for identifying the category of pixels. However, generating optimal topological structures for the HSI is a huge challenge, and

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inappropriate graph structures will reduce the model's feature learning capabilities.

Contrastive learning-based methods usually construct positive and negative sample pairs from labeled and unlabeled samples. Then, these methods aim to constrain samples from positive sample pairs closer and push samples from negative sample pairs far from each other [20], [37], [38], [39], [40], [41]. Li et al. [20] built positive and negative samples according to their similarities and designed a guided group contrastive learning to aggregate intragroup features. Cao et al. [37] constructed positive–negative sample pairs via a spatial masking enhancement strategy. Kang et al. [38] employed different enhancement strategies (e.g., spatial masking enhancement strategy, and rotation enhancement strategy) to construct positive–negative sample pairs. Liu et al. [39] pretrained the model's parameters by leveraging positive–negative sample pairs constructed from labeled samples, and the model is further used to generate pseudo-labels. However, these methods fail to construct high-quality positive–negative sample pairs when there is a small number of labeled samples, hindering the model from extracting discriminative features.

Reconstruction learning-based methods aim to achieve consistent outputs when inputs are subjected to various perturbations [22], [42], [43], [44]. Li et al. [22] proposed a diffusion-enhanced masked encoder–decoder network to extract features by simultaneously conducting denoising and reconstruction, and the encoder's parameters are finetuned using a small number of labeled samples. Jia et al. [42] designed an encoder–decoder network composed of 3-D CNNs, followed by optimizing the model's parameters through limited labeled samples. Qi et al. [43] leveraged the attention mechanism to design an encoder–decoder network, which is used to learn features by reconstructing masked spectral bands. Liu et al. [44] designed an auxiliary task to pretrain the model's weight parameters by reconstructing masked spectral information. However, these methods are prone to learning omnipotent features from numerous unlabeled samples. A small number of labeled samples struggle to effectively finetune the model's parameters, thereby achieving suboptimal classification accuracy.

Pseudo-label learning-based methods preserve unlabeled samples with confidence scores higher than a preset threshold, and predictions of these unlabeled samples corresponding to the maximum confidence score are regarded as pseudo-labels. Then, both these unlabeled samples and labeled samples are used to optimize the model's parameters [17], [18], [29], [45], [46], [47]. Yao et al. [17] used labeled samples and reliable unlabeled samples to train the model's parameters, and employed contrastive learning to leverage unreliable unlabeled samples. Wu and Prasad [18] generated pseudo-labels based on cluster labels, which are used to pretrain the model's parameters. Xu et al. [29] introduced a self-ensembling technique, which treats predictions from an ensemble network as pseudo-labels, and these pseudo-labels are further used to optimize a base network. Chen et al. [45] used unlabeled samples with pseudo-labels and labeled samples to construct a single class prototype for each category, which are further used to filter incorrect pseudo-labels. However, both spatial distribution differences and spectral variability of samples and limited labeled samples make these methods prone to overfitting. The methods that rely on confidence scores predicted by the model to produce pseudo-labels cannot guarantee the reliability of pseudo-labels, thereby reducing classification accuracy.

B. Transformer

Due to the HSI exhibiting complex spatial and spectral information, extracting rich and discriminative spatial–spectral features is crucial for accurately identifying the land-cover category of pixels. Previous studies have proven that CNN-based methods effectively extract features of the HSI through convolution, pooling, and normalization operations, achieving satisfactory classification accuracy [48], [49]. However, these methods primarily extract local information and fail to effectively capture global spatial–spectral features due to a small receptive field. The attention mechanism is excellent at modeling global dependencies, which have been widely studied in HSI classification [50], [51], [52], [53]. Zheng et al. [50] extracted spectral and spatial information via a center spectral attention module and a rectified spatial attention module. Yang et al. [51] attempted to extract spectral and spatial–spectral information through cross-spectral and cross-spatial attention modules. Wu et al. [52] designed a cooperative spectral–spatial attention module with large convolutional kernels to extract features.

Several studies utilize the vision transformer (ViT), which demonstrates powerful representation capabilities in various machine vision tasks, to extract features [17], [54]. Yao et al. [17] designed a novel position-free transformer, which decreases the impact of the spatial position distribution of hyperspectral pixels in complex scenes to extract discriminative features. Hong et al. [54] used the transformer to learn correlations between adjacent spectral bands for feature extraction. Several works propose to extract features by combining the advantages of the transformer and CNNs [14], [27], [55], [56], [57], [58]. Xu et al. [14] designed a spectral projection unit and a convolutional multihead self-attention (MHSA) unit to capture spatial and global spectral features and also developed a double branch module based on the transformer and CNNs to fuse these features. Han et al. [27] exploited both scaling factors and perturbation terms to estimate variability parameters for accurately constructing endmembers and designed a lightweight hybrid architecture consisting of the transformer and CNNs to achieve pixel-level classification. Sun et al. [55] used the transformer to mine deep semantic information from shallow spatial–spectral features extracted by 3-D CNNs. Xu et al. [56] leveraged the transformer and CNNs to extract global and local information, and these features are fused through a cross-attention mechanism module. Transformer-based methods have made significant progress in HSI classification.

III. Proposed Method

A. Overview

In this article, an RAMPL method is proposed to improve the reliability of generated pseudo-labels for accurately

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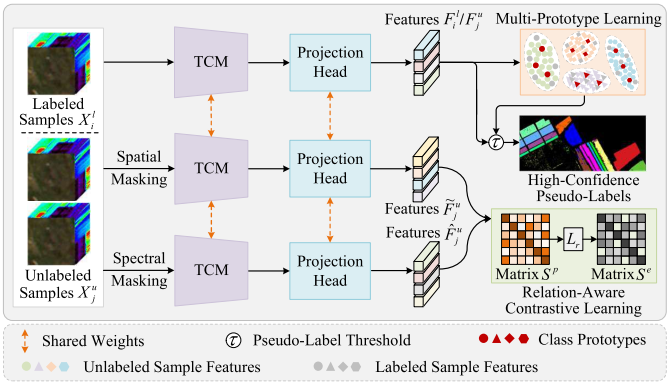


Fig. 2. Overall framework of the proposed RAMPL.

identifying the category of pixels. The overall framework of the proposed RAMPL is shown in Fig. 2. First, to effectively extract rich and discriminative spatial–spectral features, a TCM is developed to capture both global and local information. Then, to accurately generate high-confidence pseudo-labels, an MPLM is proposed to construct multiple class prototypes for each category by exploiting both unlabeled samples with pseudo-labels and a small number of labeled samples. Finally, to further enhance feature discrimination, an RACLM is built to constrain samples with the same predictions to be projected closer and push others far away in the feature space, thereby improving intraclass compactness and interclass distinguishability. Next, the detailed descriptions of TCM, MPLM, and RACLM are illustrated as follows.

B. Hybrid TCM

This article uses $\mathcal{X} \in \mathbb{R}^{H\times W\times C}$ to represent the HSI, where H, W, and C denote the height, width, and number of spectral bands, respectively. The proposed RAMPL aims to extract discriminative spatial–spectral information from both labeled pixels $\mathcal{X}_l = \{(x_i^l, y_i^l)\}_{i=1}^{N_l}$ and unlabeled pixels $\mathcal{X}_u = \{(x_j^u)\}_{j=1}^{N_u}$ for accurately identifying the category of pixels, where $y_i^l \in \mathbb{R}^D$ denotes the ground truth of the ith labeled pixel, D indicates the number of categories, $x_i^l \in \mathbb{R}^C$ and $x_j^u \in \mathbb{R}^C$ represent the ith labeled pixel and the jth unlabeled pixel, respectively. Due to pixels adjacent to the center pixel often belonging to the same category, these neighboring pixels can provide rich spatial–spectral information for the proposed RAMPL to identify the category of the center pixel. Referring to previous works [21], [22], [57], we crop HSI patches $X_i^l \in \mathbb{R}^{p\times p\times C}$ and $X_j^u \in \mathbb{R}^{p\times p\times C}$ centered on the labeled pixel x_i^l and the unlabeled pixel x_j^u as inputs, where $p \times p$ represent the spatial size of these HSI patches.

Due to the MHSA module excelling at extracting global correlations, numerous works propose various transformer-based variants as feature extractors to explore spatial–spectral information of the HSI [17], [45]. These methods usually treat pixels within HSI patches as pixel tokens and employ an MHSA module to extract global spatial–spectral features. To further enhance feature representation capabilities, they perform a nonlinear transformation via a feed-forward neural network. However, the feature extractor MHSA in these methods exhibits limitations in modeling local spatial–spectral information, while providing local context information is crucial for models to learn discriminative spatial–spectral features. To solve this problem, we propose a hybrid TCM that efficiently integrates the MHSA module with an MSCB to exploit both global and local information. The detailed structure of the proposed TCM is illustrated in Fig. 3.

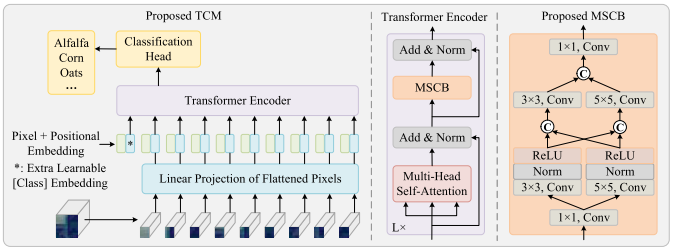


Fig. 3. Detailed structure of the proposed TCM.

Compared with the existing transformer-based variants, instead of using the standard feed-forward neural network, we design an MSCB to extract local spatial–spectral information. Specifically, as shown in Fig. 3, the output features of the MHSA module are first transformed by using a 1×1 convolutional block. Then, to effectively model local spatial–spectral correlations, two convolutional blocks with different kernel sizes, including 3×3 and 5×5, are applied to extract multiscale spatial–spectral features through multiple receptive fields. The incorporation of multiple receptive fields facilitates the proposed method to model diverse spatial–spectral correlations at different scales, thereby enhancing feature discrimination and robustness. Moreover, features from different receptive fields will interact with each other during feature extraction to improve their representation abilities. Finally, a 1×1 convolutional block is employed to refine concatenated spatial–spectral features. The working process of the proposed TCM can be described as

$$\begin{cases} z_i^{l,m} = \text{Norm}\left(\text{MHSA}\left(F_i^{l,m-1}\right) + F_i^{l,m-1}\right) \\ F_i^{l,m} = \text{Norm}\left(\text{MSCB}\left(z_i^{l,m}\right) + z_i^{l,m}\right) \end{cases} \tag{1}$$

where $F_i^{l,m}$ indicates the spatial–spectral features of the labeled sample x_i^l in the mth layer, Norm$(\cdot)$ represents the layer normalization, MHSA$(\cdot)$ denotes the multihead self-attention module, and MSCB$(\cdot)$ represents the proposed MSCB.

For labeled samples, we calculate the supervised loss of these samples through the cross-entropy loss function to optimize the proposed RAMPL's weight parameters, which can be described as

$$L_s = \frac{1}{B}\sum_{i=1}^{B} l_{\text{ce}}\left(\text{softmax}\left(F_i^l\right), y_i^l\right) \tag{2}$$

where B represents the batch size, $l_{\text{ce}}(\cdot)$ denotes the cross-entropy loss function, softmax$(\cdot)$ indicates the softmax classifier, and F_i^l and y_i^l represent the spatial–spectral features and ground truth of the labeled sample x_i^l, respectively.

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C. Multiprototype Learning Module

Existing pseudo-label learning-based methods primarily generate pseudo-labels based on confidence scores, and then unlabeled samples with pseudo-labels and labeled samples are further used to optimize the model's parameters [17], [18], [45]. However, when the number of labeled samples is small, these methods are prone to yield overconfident predictions. Moreover, the data distribution of different categories in the feature space is usually diverse, especially in complex scenarios. These methods fail to effectively model intraclass diversity when only a small number of labeled samples is available. Therefore, existing methods may generate incorrect pseudo-labels, which result in error propagation and accumulation, thus confusing model discrimination and degrading classification accuracy.

To solve this problem, an MPLM is designed, which builds multiple class prototypes for each category to generate high-confidence pseudo-labels. Due to limited labeled samples, class prototypes constructed using only a few labeled samples may suffer from confirmation bias and struggle to model the complex data distribution of the HSI. Instead of using only a small number of labeled samples, the proposed MPLM exploits spatial–spectral features of unlabeled samples with pseudo-labels and labeled samples to construct multiple class prototypes for each category using the K-means clustering algorithm [59], [60], which can effectively model intraclass diversity. The class prototypes of the dth category at the tth iteration can be expressed as

$$P_d^k = \text{K-means}\left(\left(F_i^{l,t}, y_i^l\right), \left(F_j^{u,t\text{-}1}, \hat{y}_j^{u,t\text{-}1}\right)\right), \quad k \in \{1, \ldots, K\} \tag{3}$$

where K-means$(\cdot)$ denotes the K-means clustering algorithm, K represents the number of class prototypes for each category and is set to 3 in experiments, $F_i^{l,t}$ and y_i^l denote the spatial–spectral features and ground truth of the labeled sample x_i^l at the tth iteration, and $F_j^{u,t\text{-}1}$ and $\hat{y}_j^{u,t\text{-}1}$ represent the spatial–spectral features and pseudo-label of the unlabeled sample x_j^u at the $(t\text{-}1)$th iteration, respectively.

To efficiently utilize massive unlabeled samples, the proposed RAMPL generates a high-confidence pseudo-label set $\mathcal{D}_p$ based on the spatial–spectral features of unlabeled samples and the class prototype set. Specifically, if an unlabeled sample significantly deviates from all class prototypes in the feature space, this sample is directly discarded. Otherwise, it is preserved. To measure the deviation between unlabeled samples and the class prototype set, the minimum distances between them are first calculated, which can be described as

$$D_j^u = \text{Min}\left(\text{Dis}\left(F_j^u, P\right)\right) \tag{4}$$

where Min$(\cdot)$ represents the minimum function, Dis$(\cdot)$ is the Euclidean distance function, F_j^u is the spatial–spectral features of the unlabeled sample x_j^u, $P = \{P_1^k, \ldots, P_d^k, \ldots, P_D^k\}$ denotes the class prototype set, D indicates the number of categories, $P_d^k \in \mathbb{R}^{K \times Dim}$ represents the class prototypes of the dth category defined in (3), and Dim denotes the feature dimension.

Then, the unlabeled sample x_j^u whose distance D_j^u belongs to the bottom pseudo-label ratio γ is preserved, and the distance corresponding to the γ is regarded as the pseudo-label threshold τ. For the preserved unlabeled sample x_j^u, the proposed RAMPL treats the index corresponding to the minimum distance as a high-confidence pseudo-label $\hat{y}_j^u$, which can be described as

$$\hat{y}_j^u = \begin{cases} \dfrac{\text{Index}\left(\text{Dis}\left(F_j^u, P\right)\right)}{K}, & \text{if } D_j^u \le \tau \\ \text{ignore}, & \text{otherwise} \end{cases} \tag{5}$$

where Index$(\cdot)$ represents the index calculation function of the minimum distance.

Finally, these preserved unlabeled samples are further used to optimize the proposed RAMPL's weight parameters. Therefore, the supervised loss calculated from these preserved unlabeled samples can be described as

$$L_p = \frac{1}{\left|\mathcal{D}_p\right|} \sum_{j=1}^{\mu B} \mathbb{1}\left(D_j^u \le \tau\right) \cdot l_{\text{ce}}\left(\text{softmax}\left(F_j^u\right), \hat{y}_j^u\right) \tag{6}$$

where $\mathbb{1}(\cdot)$ represents the indicator function, B denotes the batch size, μ represents the relative ratio between unlabeled and labeled samples, $l_{\text{ce}}(\cdot)$ indicates the cross-entropy loss function, and softmax$(\cdot)$ denotes the softmax classifier.

D. Relation-Aware Contrastive Learning Module

To utilize spatial–spectral information of unlabeled samples, existing pseudo-label learning methods typically focus on generating high-confidence pseudo-labels for unlabeled samples and incorporate them into model training [17], [18], [45]. Since these methods struggle to effectively learn the potential differences between samples in spatial distribution and spectral bands, they fail to extract discriminative spatial–spectral features when labeled samples are insufficient, thereby reducing the reliability of generated pseudo-labels. To address this problem, an RACLM is introduced to learn the complex spatial distribution difference and spectral variability of the HSI by building a relation-aware pseudo-graph [61], [62], thereby improving the model discrimination. The proposed relation-aware pseudo-graph consists of a pseudo-relation matrix and an embedding relation matrix. Specifically, to effectively model potential relations between unlabeled samples, the proposed RAMPL first constructs the pseudo-relation matrix $S^p \in \mathbb{R}^{\mu B \times \mu B}$ by leveraging their confidence scores, which can be described as

$$S_{b,j}^p = \begin{cases} 1, & \text{if } b = j \\ p_b \cdot p_j, & \text{if } b \ne j \text{ and } \left\{x_b^u, x_j^u\right\} \in \mathcal{D}_p \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where $p_b, p_j \in \mathbb{R}^D$ represent the confidence scores of unlabeled samples x_b^u and x_j^u with pseudo-labels in the high-confidence pseudo-label set $\mathcal{D}_p$.

To construct the embedding relation matrix $S^e \in \mathbb{R}^{\mu B \times \mu B}$, the proposed RAMPL then employs different data augmentation strategies $\tilde{T}$ and $\hat{T}$ (spatial and spectral masking augmentation strategies are applied) to build the sample pair $(\tilde{x}_j^u, \hat{x}_j^u)$ from the same unlabeled sample x_j^u. The sample pair is fed into the proposed TCM, followed by

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a projection head consisting of two fully connected layers to obtain spatial–spectral features ($\tilde{F}_j^u$, $\hat{F}_j^u$). The embedding relation matrix S^e is constructed from spatial–spectral features ($\tilde{F}_j^u$, $\hat{F}_j^u$) of the sample pair ($\tilde{x}_j^u$, $\hat{x}_j^u$), which can be described as

$$S_{b,j}^e = \begin{cases} \exp\left(\mathrm{PrH}\left(\tilde{F}_b^u\right) \cdot \mathrm{PrH}\left(\hat{F}_j^u\right)\right), & \text{if } b = j \\ \exp\left(\mathrm{PrH}\left(\tilde{F}_b^u\right) \cdot \mathrm{PrH}\left(\tilde{F}_j^u\right)\right), & \text{otherwise} \end{cases} \tag{8}$$

where $\exp(\cdot)$ represents the exponential function and $\mathrm{PrH}(\cdot)$ denotes the projection head.

To improve spatial–spectral feature discrimination and robustness, a relation-aware graph contrastive loss is proposed, which aims to constrain unlabeled samples with the same predictions to learn consistent embeddings and push others far away in the feature space by minimizing differences between the two normalized relation matrices. In this way, the proposed method can effectively learn potential differences between samples in spatial distribution and spectral bands. The proposed relation-aware graph contrastive loss can be described as

$$\begin{cases} L_r = \dfrac{1}{\mu B} \displaystyle\sum_{m=1}^{\mu B} \sum_{n=1}^{\mu B} M_{m,n} \cdot \left\| S_{m,n}^p - S_{m,n}^e \right\|_2 \\ M_{m,n} = \mathbb{1}\left[(p_m \cdot p_n > T) \| (p_m \cdot p_n < 1 - T) \right] \end{cases} \tag{9}$$

where B indicate the batch size, μ denotes the relative ratio between unlabeled and labeled samples, $\|\cdot\|_2$ represents the l_2 regularization function, $\mathbb{1}(\cdot)$ denotes the indicator function, $p_m, p_n \in \mathbb{R}^D$ represent the confidence scores of unlabeled samples x_m^u and x_n^u, and T is the pseudo-relation threshold.

In addition, neighboring pixels often exhibit similar spectral and spatial information within HSI patches. To improve the model's capabilities in extracting local spatial and spectral correlations, the proposed RAMPL reconstructs the original input tokens using a projection head consisting of two fully connected layers and uses the mean square error function to calculate spatial masking loss L_{spa} and spectral masking loss L_{spe} to further optimize the model's parameters.

Based on the above analysis, the overall loss of the proposed RAMPL can be described as follows:

$$L = L_s + \lambda_p L_p + \lambda_r L_r + \lambda_{\mathrm{spa}} L_{\mathrm{spa}} + \lambda_{\mathrm{spe}} L_{\mathrm{spe}} \tag{10}$$

where λ_p, λ_r, λ_{spa}, and λ_{spe} represent the tradeoff coefficients of different loss terms.

IV. Experiments

A. Datasets and Compared Methods

To evaluate the classification accuracy of the proposed RAMPL, this article utilizes three common evaluation metrics, including overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. A larger value of these evaluation metrics represents higher performance. Moreover, we conduct extensive experiments on the Indian Pines (IP), Houston2013 (H2013), Salinas (SA), and Kennedy Space Center (KSC) datasets. To quantitatively and qualitatively analyze the classification accuracy of the proposed RAMPL, 11 state-of-the-art methods, including SSFTT [55], DBCTNet [14], S^{2}VNet [27], AMGCFN [30], SSLSM [44], PUSL [17], DSCA-Net [21], DEMAE [22], HGCN-MHF [9], RSEN [29], and PPLR [45], are used as compared methods, and the experimental settings of these compared methods follow their original papers.

B. Parameter Settings

To verify the effectiveness of the proposed RAMPL, we randomly select five hyperspectral pixels in each category as labeled samples and the remaining pixels as unlabeled samples. The batch size is set to 512 across all experiments, and a fixed learning rate of 0.0001 is set during the model training process. The relative ratio μ between unlabeled and labeled samples is set to 2 in experiments. The number of epochs is set to 700 on the IP, H2013, SA, and KSC datasets. Moreover, the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$ is adopted to optimize our proposed RAMPL's weight parameters. The proposed TCM employs $L = 4$ layers of transformer encoder for feature extraction. The feature dimension of the proposed TCM is set to 128, and the MHSA module contains four parallel heads. In this article, the K-means clustering algorithm with 300 iterations is applied to construct multiple class prototypes for each category. All experiments, including the proposed RAMPL and other compared methods, are conducted ten times, and the average value of these experimental results is treated as the final classification performance. In addition, the proposed RAMPL is implemented using the PyTorch framework with version 1.10, and all experiments are conducted on an Ubuntu 20.04 workstation with NVIDIA 3090 GPUs.

C. Classification Performance

1) IP Dataset: To evaluate the classification accuracy of RAMPL, we first conduct extensive experiments on the IP dataset, and the results with five labeled samples in each category are reported in Table I. It can be observed that the proposed RAMPL achieves the best classification accuracy and significantly outperforms other state-of-the-art methods. A small number of labeled samples results in transformer-based methods failing to learn discriminative features, thereby restricting classification accuracy. The OAs of SSFTT, DBCTNet, and S^{2}VNet are only 58.17%, 63.37%, and 67.19%, respectively. PUSL struggles to generate high-confidence pseudo-labels under limited labeled samples, which struggles to achieve excellent classification accuracy. The OA, AA, and Kappa of PUSL are 70.24%, 79.13%, and 66.34%, respectively. PPLR uses unlabeled samples with pseudo-labels and labeled samples to construct a single class prototype for each category, which is further used to filter out incorrect pseudo-labels. However, the single-class prototype fails to capture intraclass diversity in complex scenes. The classification accuracy of PPLR is limited under a few labeled samples, and the OA is 69.28%. To generate high-confidence pseudo-labels, RAMPL uses unlabeled samples with pseudo-labels and labeled samples to construct multiple class prototypes for each category, which effectively models intraclass diversity, thereby achieving the best classification accuracy. In terms of OA, compared with SSFTT, DBCTNet, S^{2}VNet, AMGCFN, SSLSM, PUSL, DSCA-Net, DEMAE, HGCN-MHF, RSEN,

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TABLE I
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS WITH FIVE LABELED SAMPLES PER CATEGORY ON THE IP DATASET

Category	SSFTT	DBCTNet	S^2VNet	AMGCFN	SSLSM	PUSL	DSCA-Net	DEMAE	HGCN-MHF	RSEN	PPLR	RAMPL
1	88.29	98.78	85.37	98.78	99.35	80.00	99.51	**100.0**	97.91	84.96	96.09	80.43
2	38.22	42.78	49.38	52.45	54.72	53.89	63.15	47.32	63.82	**76.10**	58.11	63.24
3	43.58	52.18	42.34	66.10	59.22	56.07	56.73	60.24	74.25	11.56	45.40	**95.54**
4	61.72	86.81	75.22	96.33	84.90	89.62	96.73	93.45	92.24	24.28	79.37	**98.73**
5	77.20	73.31	68.38	75.19	63.50	72.63	71.46	79.21	**80.42**	15.83	72.53	63.56
6	90.11	91.05	93.61	84.66	87.69	86.22	86.90	92.00	96.05	**98.18**	92.89	96.71
7	98.26	**100.0**	97.83	98.64	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
8	77.38	80.34	78.75	96.31	95.31	91.92	**100.0**	96.26	99.71	**100.0**	95.36	99.79
9	98.67	94.67	**100.0**	**100.0**	**100.0**	96.00	**100.0**	98.00	**100.0**	**100.0**	99.00	**100.0**
10	51.43	54.19	65.55	70.73	64.45	66.93	71.54	58.00	69.57	16.96	63.00	**84.47**
11	51.85	51.54	64.72	69.84	64.16	65.82	77.13	54.43	82.33	70.83	61.18	**88.31**
12	33.01	48.83	47.56	57.32	55.01	54.50	50.92	53.08	71.88	18.57	54.23	**88.87**
13	98.80	97.45	96.50	98.45	93.22	90.15	98.10	97.15	99.33	99.50	99.32	**100.0**
14	75.48	84.60	92.10	86.15	86.51	85.79	89.97	91.79	88.84	**97.00**	85.56	96.05
15	54.75	82.10	57.04	89.82	88.86	76.58	92.70	86.77	**94.87**	33.33	81.79	66.58
16	99.66	**100.0**	97.54	96.69	97.31	**100.0**	99.32	96.70	99.66	92.80	99.35	98.92
OA	58.17	63.37	67.19	73.40	70.42	70.24	76.37	68.24	81.12	61.03	69.28	**85.76**
AA	71.15	77.41	75.74	83.59	80.89	79.13	84.63	81.52	88.31	64.99	80.20	**88.83**
Kappa	53.09	59.11	62.80	69.99	66.79	66.34	73.44	64.52	78.62	54.49	65.27	**83.77**

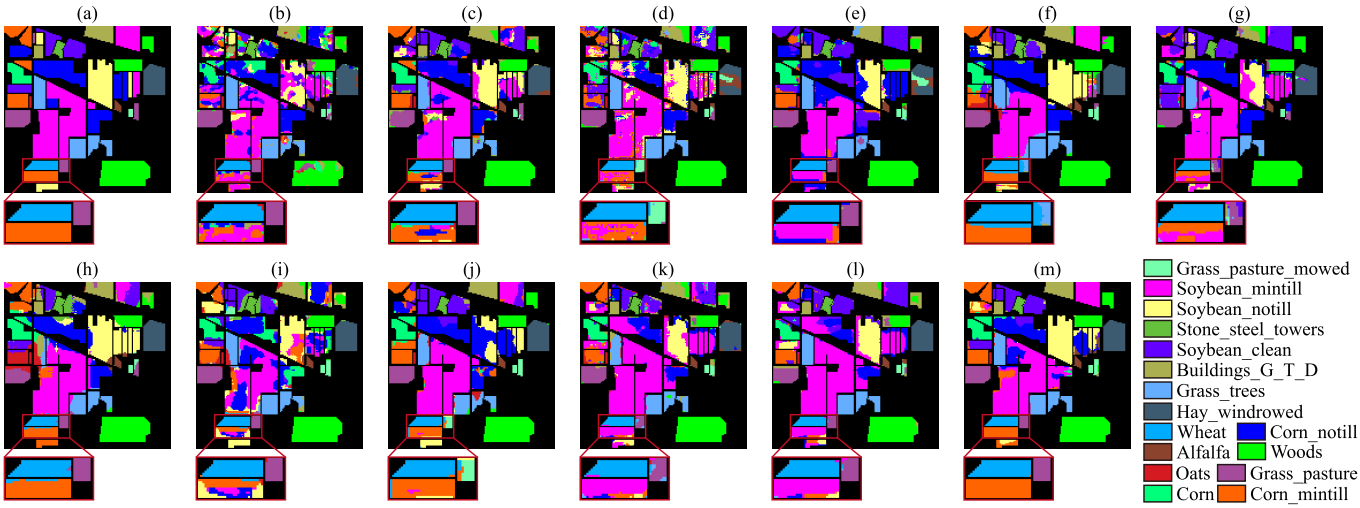


Fig. 4. Classification maps of different methods with five labeled samples per category on the IP dataset, including (a) ground truth, (b) SSFTT, (c) DBCTNet, (d) S^2VNet, (e) AMGCFN, (f) SSLSM, (g) PUSL, (h) DSCA-Net, (i) DEMAE, (j) HGCN-MHF, (k) RSEN, (l) PPLR, and (m) RAMPL.

and PPLR, the proposed RAMPL improves 27.59%, 22.39%, 18.57%, 12.36%, 15.34%, 15.52%, 9.39%, 17.52%, 4.64%, 24.73%, and 16.48%, respectively.

The classification maps of the proposed RAMPL and the compared methods on the IP dataset are illustrated in Fig. 4. Due to different categories exhibiting complex boundary information, SSFTT, DBCTNet, and S^2VNet cannot extract discriminative features under limited labeled samples, resulting in these methods failing to effectively identify the land-cover category of pixels in the zoomed-in-view area. AMGCFN, DSCA-Net, and HGCN-MHF struggle to build optimal topological structures, which fail to achieve satisfactory performance in the "Corn_mintill" and "Grass_pasture" categories. The proposed RAMPL uses TCM to extract discriminative features and constructs multiple class prototypes for each category using unlabeled samples with pseudo-labels and labeled samples to generate high-quality pseudo-labels, thereby achieving excellent classification maps.

2) H2013 Dataset: We further demonstrate the effectiveness of the proposed RAMPL on the H2013 dataset, and the results with five labeled samples in each category are listed in Table II. We observe that RAMPL achieves the best performance on different metrics. Due to the H2013 dataset exhibiting complex spectral and spatial information, graph learning-based methods, which treat small areas as nodes, are easily affected by background pixels. For example, the OAs of AMGCFN, DSCA-Net, and HGCN-MHF are 75.69%, 79.18%, and 77.16%, respectively. SSLSM and DEMAE utilize unlabeled samples to pretrain the model's parameters, followed by finetuning with labeled samples. However, limited labeled samples fail to provide sufficient supervised signals, achieving suboptimal classification performance. The OAs of SSLSM and DEMAE are 74.53% and 79.37%, respectively. RAMPL employs TCM to extract both global and local information and leverages RACLM to learn the complex spatial distribution difference and spectral

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TABLE II
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS WITH FIVE LABELED SAMPLES PER CATEGORY ON THE H2013 DATASET

Category	SSFTT	DBCTNet	S^2VNet	AMGCFN	SSLSM	PUSL	DSCA-Net	DEMAE	HGCN-MHF	RSEN	PPLR	RAMPL
1	82.50	90.93	98.07	80.25	66.13	87.31	84.25	80.90	87.02	95.21	87.68	**98.56**
2	85.52	69.30	86.19	68.28	67.66	81.33	83.80	85.16	69.64	97.99	84.87	**99.60**
3	98.45	99.57	99.42	98.31	95.65	95.28	97.14	97.23	**99.90**	73.00	94.33	99.71
4	85.70	**94.87**	92.82	81.52	79.23	89.79	82.11	86.49	85.81	83.43	92.17	93.41
5	97.13	95.13	94.62	94.27	88.74	98.00	95.97	98.92	95.16	**100.0**	99.07	**100.0**
6	90.19	85.86	76.09	79.80	89.66	88.95	85.13	**93.38**	87.28	75.57	86.77	90.15
7	65.39	87.93	67.46	77.66	81.96	74.82	74.01	72.66	79.69	**90.54**	77.62	74.76
8	40.33	44.03	37.41	44.50	**58.61**	46.51	47.80	51.19	40.21	38.36	46.95	52.49
9	41.41	81.42	77.39	67.35	62.21	76.54	75.78	73.08	69.25	74.04	75.22	**88.58**
10	59.57	73.96	53.72	63.80	59.84	65.26	85.74	76.51	71.30	82.16	62.21	**95.68**
11	72.92	71.40	53.78	**81.36**	77.41	53.57	72.05	78.46	77.76	77.37	59.28	16.68
12	75.98	53.26	67.59	63.22	61.87	71.48	62.61	56.50	55.48	60.76	67.59	**95.86**
13	85.00	88.74	81.47	80.05	86.87	93.41	92.16	87.52	85.26	48.02	94.48	**94.88**
14	95.44	99.82	98.11	98.37	99.98	97.71	99.95	**100.0**	**100.0**	99.76	98.57	**100.0**
15	97.80	99.77	**100.0**	97.40	97.55	96.98	86.99	98.38	**100.0**	38.35	96.88	**100.0**
OA	74.68	79.64	76.41	75.69	74.53	78.02	79.18	79.37	77.16	77.36	78.65	**84.36**
AA	78.22	82.40	78.94	78.41	78.22	81.13	81.70	82.42	80.37	75.64	81.58	**86.69**
Kappa	72.64	78.00	74.54	73.75	72.49	76.27	77.51	77.72	75.34	75.57	76.95	**83.11**

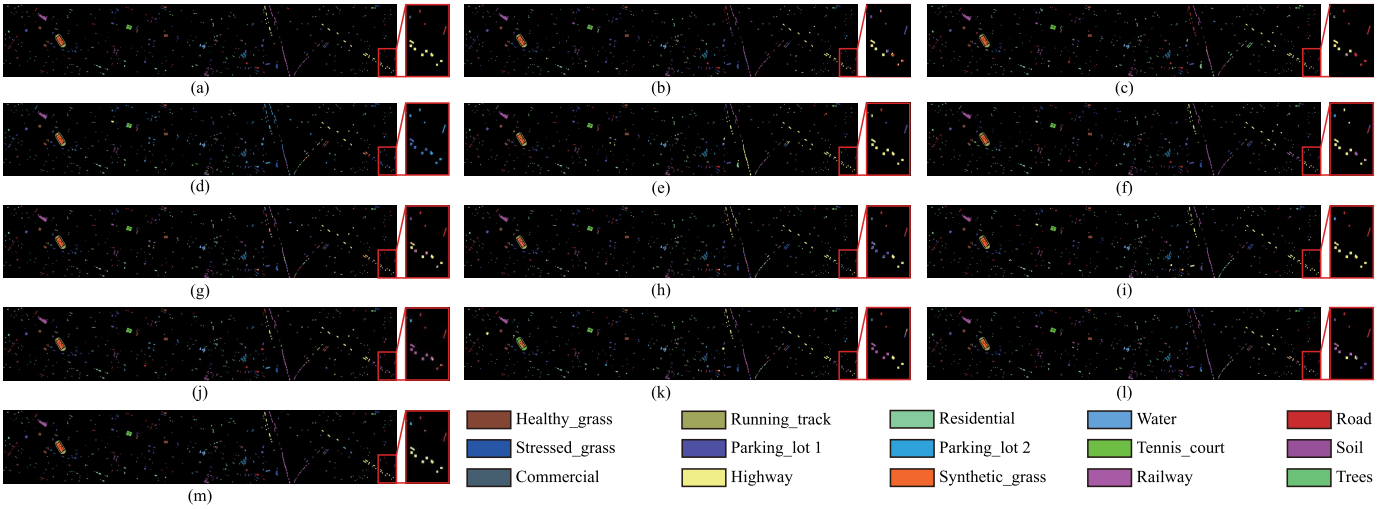


Fig. 5. Classification maps of different methods with five labeled samples per category on the H2013 dataset, including (a) ground truth, (b) SSFTT, (c) DBCTNet, (d) S^2VNet, (e) AMGCFN, (f) SSLSM, (g) PUSL, (h) DSCA-Net, (i) DEMAE, (j) HGCN-MHF, (k) RSEN, (l) PPLR, and (m) RAMPL.

variability, improving the reliability of pseudo-labels and achieving superior performance. In terms of OA, compared with SSFTT, DBCTNet, S^2VNet, AMGCFN, SSLSM, PUSL, DSCA-Net, DEMAE, HGCN-MHF, RSEN, and PPLR, the proposed RAMPL improves 9.68%, 4.72%, 7.95%, 8.67%, 9.83%, 6.34%, 5.18%, 4.99%, 7.20%, 7.00%, and 5.71%, respectively.

The classification maps of the proposed RAMPL and the compared methods on the H2013 dataset are illustrated in Fig. 5. Compared with the IP dataset, the H2013 dataset exhibits complex spatial and spectral information. Compared methods, such as DBCTNet, S^2VNet, SSLSM, DEMAE, RSEN, and PPLR, are prone to being disturbed by background pixels and cannot extract discriminative spatial–spectral features under limited labeled samples. These methods have significant mispredictions in the classification maps. The proposed RAMPL uses TCM to extract both global and local features, achieving the best classification map. For example, the proposed RAMPL can accurately identify the "Highway" and "Road" categories in the zoomed-in-view area.

3) SA Dataset: Similar to the IP and H2013 datasets, we conduct extensive experiments on the SA dataset to discuss the classification accuracy of RAMPL. Table III lists the experimental results of RAMPL and different compared methods with 5 labeled samples in each category. Limited labeled samples result in compared methods, including PUSL, RSEN, and PPLR, struggling to extract discriminative features. Therefore, these methods are unable to produce high-confidence pseudo-labels. Incorrect predictions in pseudo-labels will confuse the classification network, thus reducing classification accuracy. The OAs of PUSL, RSEN, and PPLR are only 87.85%, 84.72%, and 88.05%, respectively. Since samples with the same category are compactly distributed on the SA dataset, graph-based methods such as AMGCFN, DSCA-Net, and HGCN-MHF effectively extract discriminative spatial–spectral features. The OAs of AMGCFN, DSCA-Net, and HGCN-MHF are 91.34%,

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TABLE III
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS WITH FIVE LABELED SAMPLES PER CATEGORY ON THE SA DATASET

Category	SSFTT	DBCTNet	S^2VNet	AMGCFN	SSLSM	PUSL	DSCA-Net	DEMAE	HGCN-MHF	RSEN	PPLR	RAMPL
1	99.86	99.10	93.31	96.71	98.46	95.73	**100.0**	99.55	**100.0**	99.23	95.83	99.83
2	99.29	99.73	98.14	98.98	99.84	98.60	**100.0**	99.98	**100.0**	95.16	99.83	99.08
3	95.80	84.02	65.28	95.88	99.70	83.81	**100.0**	**100.0**	**100.0**	94.63	87.47	98.89
4	98.72	98.98	99.81	97.09	97.99	95.65	**100.0**	97.98	**100.0**	98.84	98.63	98.82
5	97.19	92.38	92.77	92.04	96.90	93.02	73.18	92.83	84.55	**99.51**	91.56	96.70
6	99.24	99.32	99.74	97.53	**100.0**	99.98	99.80	97.91	**100.0**	**100.0**	99.79	**100.0**
7	99.87	99.55	99.54	98.41	96.00	95.39	99.97	99.65	**100.0**	99.64	98.96	98.88
8	78.27	59.62	71.32	75.88	71.78	74.29	**87.94**	77.02	80.22	35.28	72.10	84.85
9	99.98	99.12	99.71	99.91	**100.0**	97.44	**100.0**	99.99	**100.0**	99.12	99.44	99.85
10	91.90	87.31	74.24	89.58	94.17	86.20	**98.08**	93.79	96.36	95.50	89.73	93.52
11	98.19	95.97	82.93	98.73	78.93	94.13	97.18	**99.93**	99.72	99.35	94.21	98.13
12	98.40	94.91	99.26	99.41	98.65	97.63	90.27	97.24	**100.0**	97.27	98.55	99.96
13	95.46	99.28	96.86	97.92	**100.0**	99.31	99.56	98.51	**100.0**	99.98	99.43	99.97
14	93.31	94.48	92.15	98.31	65.70	93.60	98.59	95.00	**98.97**	98.38	96.35	98.22
15	69.95	78.34	60.54	87.18	87.63	73.71	87.13	89.83	85.58	**96.73**	70.79	79.37
16	93.37	93.74	96.56	98.15	98.06	94.44	86.46	98.65	**99.94**	94.54	96.15	97.69
OA	89.97	86.10	84.39	91.34	90.35	87.85	**93.41**	92.60	92.93	84.72	88.05	93.14
AA	94.30	92.24	88.88	95.11	92.74	92.06	94.88	96.12	**96.58**	93.95	93.05	96.48
Kappa	88.85	84.61	82.62	90.39	89.30	86.52	92.18	91.79	92.14	83.18	86.72	**92.36**

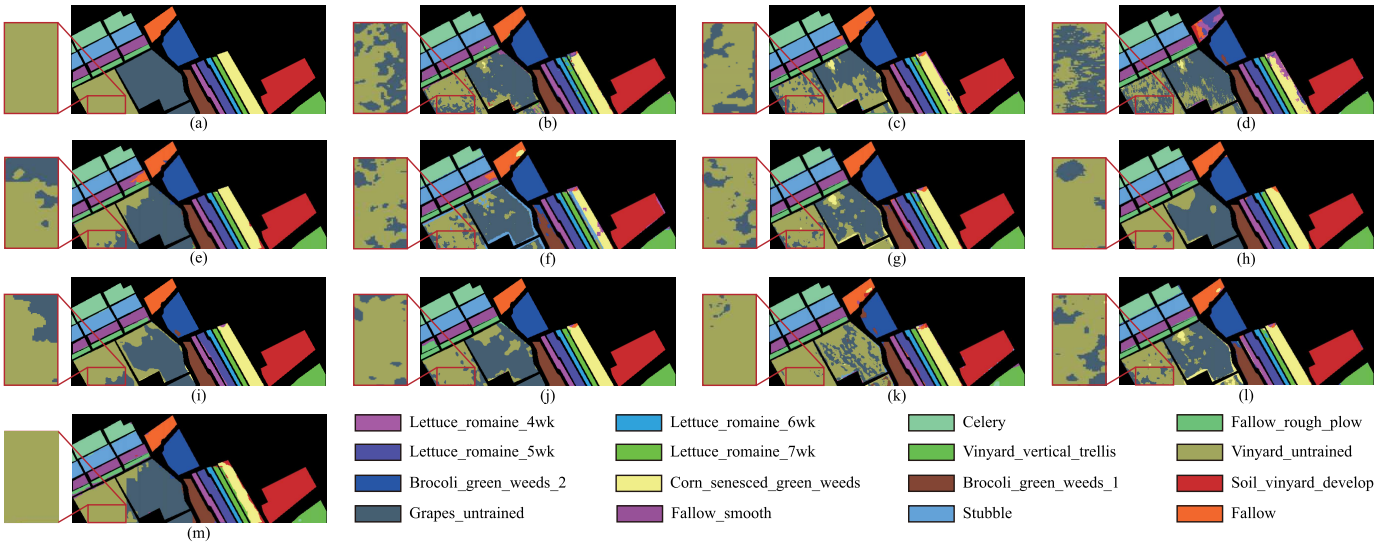


Fig. 6. Classification maps of different methods with five labeled samples per category on the SA dataset, including (a) ground truth, (b) SSFTT, (c) DBCTNet, (d) S^2VNet, (e) AMGCFN, (f) SSLSM, (g) PUSL, (h) DSCA-Net, (i) DEMAE, (j) HGCN-MHF, (k) RSEN, (l) PPLR, and (m) RAMPL.

93.41%, and 92.93%, respectively. RAMPL introduces MPLM to generate high-confidence pseudo-labels and improves the reliability of pseudo-labels through RACLM, thereby improving classification accuracy. In terms of Kappa, compared with SSFTT, DBCTNet, S^2VNet, AMGCFN, SSLSM, PUSL, DSCA-Net, DEMAE, HGCN-MHF, RSEN, and PPLR, the proposed RAMPL improves 3.51%, 7.75%, 9.74%, 1.97%, 3.06%, 5.84%, 0.18%, 0.57%, 0.22%, 9.18%, and 5.64%, respectively.

The classification maps of the proposed RAMPL and the compared methods on the SA dataset are illustrated in Fig. 6. The proposed RAMPL constructs multiple class prototypes for each category to model intraclass diversity and uses RACLM to learn the complex spatial distribution difference and spectral variability, which improves feature discrimination and achieves excellent classification performance. The proposed RAMPL achieves smoother classification maps compared with other methods in the zoomed-in-view area.

4) KSC Dataset: We conduct extensive experiments to explore the performance of RAMPL and other methods on the KSC dataset. Table IV reports the experimental results of different methods with five labeled samples in each category. SSFTT, DBCTNet, and S^2VNet do not achieve satisfactory classification accuracy under limited labeled samples, and the OAs of these methods are 71.41%, 87.51%, and 88.96%, respectively. PUSL and PPLR leverage confidence scores to selectively preserve unlabeled samples and generate corresponding pseudo-labels. These unlabeled samples and limited labeled samples are further utilized to optimize the model's parameters. The OAs of PUSL and PPLR are 88.31% and 89.85%, respectively. Due to samples from the same category being dispersed in the space domain, AMGCFN, DSCA-Net, and HGCN-MHF effectively extract discriminative features on the KSC dataset. The OAs of AMGCFN, DSCA-Net, and HGCN-MHF are 93.63%, 94.35%, and 93.35%, respectively. The proposed RAMPL uses unlabeled samples

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TABLE IV
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS WITH FIVE LABELED SAMPLES PER CATEGORY ON THE KSC DATASET

Category	SSFTT	DBCTNet	S^2VNet	AMGCFN	SSLSM	PUSL	DSCA-Net	DEMAE	HGCN-MHF	RSEN	PPLR	RAMPL
1	70.64	86.67	95.81	89.92	96.29	95.66	**100.0**	93.41	98.63	85.01	98.03	99.20
2	56.43	87.98	78.54	92.02	24.27	83.95	80.25	81.08	70.27	15.29	**97.94**	79.15
3	38.69	90.24	79.15	**100.0**	99.60	82.03	78.49	99.60	99.73	89.08	65.63	94.76
4	56.88	55.95	52.71	54.88	80.57	44.05	**99.60**	82.67	74.71	37.57	80.16	66.67
5	47.79	72.37	74.53	72.44	86.62	41.61	80.13	77.14	74.52	68.65	72.67	**95.39**
6	24.37	73.30	90.15	**100.0**	98.22	92.14	**100.0**	**100.0**	99.41	27.01	80.79	**100.0**
7	61.76	94.10	99.75	**100.0**	**100.0**	88.57	**100.0**	**100.0**	**100.0**	47.10	92.38	97.94
8	78.36	78.69	83.76	97.64	**100.0**	72.62	**100.0**	70.93	85.95	50.70	82.37	**100.0**
9	87.91	89.69	86.22	92.61	83.69	94.23	84.24	97.54	89.19	88.91	83.08	**100.0**
10	80.99	93.41	88.22	**99.25**	85.93	97.77	81.11	96.57	95.31	47.12	90.10	91.16
11	79.89	97.66	91.17	**100.0**	**100.0**	99.28	99.76	**100.0**	95.48	22.29	99.05	99.02
12	53.94	81.85	94.05	94.56	98.99	86.88	**100.0**	88.52	99.40	22.03	86.68	**100.0**
13	93.70	99.91	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	88.26	**100.0**	**100.0**
OA	71.41	87.51	88.96	93.63	91.68	88.31	94.35	92.58	93.35	60.01	89.85	**96.13**
AA	63.95	84.76	85.70	91.79	88.78	82.99	92.58	91.34	90.97	53.00	86.84	**94.10**
Kappa	68.16	86.12	87.71	92.92	90.74	86.99	93.71	91.74	92.60	55.10	88.70	**95.69**

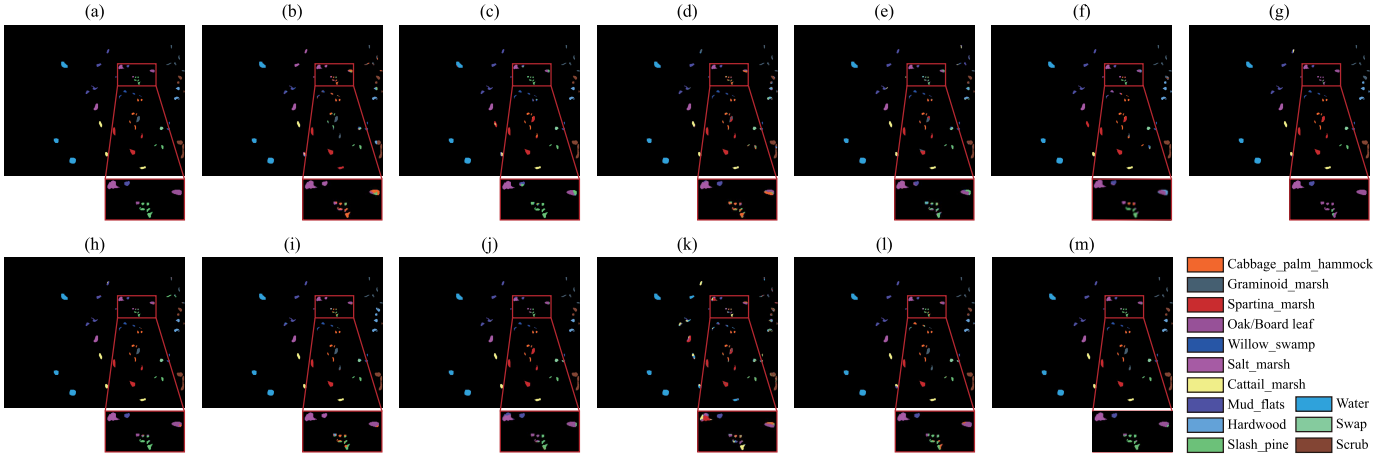


Fig. 7. Classification maps of different methods with five labeled samples per category on the KSC dataset, including (a) ground truth, (b) SSFTT, (c) DBCTNet, (d) S^2VNet, (e) AMGCFN, (f) SSLSM, (g) PUSL, (h) DSCA-Net, (i) DEMAE, (j) HGCN-MHF, (k) RSEN, (l) PPLR, and (m) RAMPL.

with pseudo-labels and labeled samples to construct multiple class prototypes for generating high-quality pseudo-labels and leverages RACLM to learn the complex spatial distribution difference and spectral variability, improving the reliability of pseudo-labels. Therefore, the proposed RAMPL achieves the best performance on three evaluation metrics. In terms of OA, compared with SSFTT, DBCTNet, S^2VNet, AMGCFN, SSLSM, PUSL, DSCA-Net, DEMAE, HGCN-MHF, RSEN, and PPLR, the proposed RAMPL improves 24.72%, 8.62%, 7.17%, 2.50%, 4.45%, 7.82%, 1.78%, 3.55%, 2.78%, 36.12%, and 6.28%, respectively.

The classification maps of the proposed RAMPL and the compared methods on the KSC dataset are illustrated in Fig. 7. SSFTT, PUSL, and PPLR primarily use the transformer to extract global features, which cannot extract local spatial–spectral information, reducing classification performance. RSEN focuses on a fixed receptive field, which cannot extract diverse local information, limiting feature discrimination. RAMPL extracts global and diverse local information via the proposed TCM and uses RACLM to learn the complex spatial distribution difference and spectral variability. Similar to the IP, H2013, and SA datasets, RAMPL achieves excellent classification maps.

TABLE V
IMPACT OF THE PROPOSED TCM ON CLASSIFICATION PERFORMANCE (OA: %)

Method	IP	H2013	SA	KSC
TCM w/o MHSA	61.30	70.07	80.63	81.69
TCM w/o MSCB	62.56	71.11	81.37	82.76
TCM	64.42	72.78	83.59	85.02
RAMPL w/o MSCB	81.67	78.41	90.02	94.17
RAMPL	85.76	84.36	93.14	96.13

D. Ablation Study

1) Impact of Proposed TCM: To analyze the impact of the proposed TCM on classification accuracy, we perform extensive experiments on the IP, H2013, SA, and KSC datasets, and the experimental results with five labeled samples in each category are reported in Table V. TCM w/o MHSA, TCM w/o MSCB, and TCM optimize the model's weight parameters

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TABLE VI
IMPACT OF THE PROPOSED MPLM ON CLASSIFICATION PERFORMANCE (OA: %)

Method	IP	H2013	SA	KSC
RAMPL w/o MPLM^{-}	62.43	67.44	84.53	85.97
RAMPL w/o MPLM	72.48	75.54	88.21	90.79
RAMPL w/o MPLM*	61.57	68.17	84.42	83.07
RAMPL	85.76	84.36	93.14	96.13

TABLE VII
IMPACT OF THE PROPOSED RACLM ON CLASSIFICATION PERFORMANCE (OA: %)

Method	IP	H2013	SA	KSC
RAMPL w/o RACLM	82.12	81.61	91.38	94.95
RAMPL	85.76	84.36	93.14	96.13

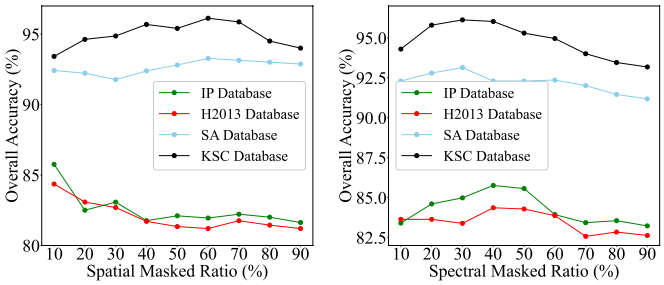


Fig. 8. Classification accuracy with different masking ratios in the spatial and spectral dimensions.

in a supervised manner, where TCM w/o MHSA represents that only using the proposed MSCB as the feature extractor, TCM w/o MSCB represents that three key modules in the proposed RAMPL, including MSCB, MPLM, and RACLM, are removed and only MHSA is used to extract features, and TCM denotes that combining both MSCB and MHSA for feature extraction. RAMPL w/o MSCB represents removing MSCB and only using MHSA to extract features from labeled and unlabeled samples. As shown in Table V, due to the proposed RAMPL fully leveraging unlabeled samples, the classification performance of the proposed RAMPL significantly improves compared with TCM. Moreover, the proposed TCM efficiently integrates the advantages of both MHSA and MSCB to extract global and local spatial–spectral features, thereby improving feature discrimination and classification performance. It can be observed that when MHSA and MSCB work simultaneously, the proposed RAMPL achieves the best performance compared to RAMPL w/o MSCB. In terms of OA, the proposed RAMPL improves 4.09%, 5.95%, 3.12%, and 1.96% on the IP, H2013, SA, and KSC datasets, respectively.

2) Impact of Proposed MPLM: To verify the effectiveness of the proposed MPLM, we conduct extensive experiments on the IP, H2013, SA, and KSC datasets. The results are listed in Table VI, where RAMPL w/o MPLM represents that the K-means clustering algorithm is applied to generate a single class prototype for each category using spatial–spectral features of unlabeled samples with pseudo-labels and labeled samples, RAMPL w/o MPLM^{-} means that the channelwise average value of spatial–spectral features of labeled samples belonging to the same category is treated as the class prototype for each category, and RAMPL w/o MPLM* indicates that high-confidence pseudo-labels are generated based on confidence scores of unlabeled samples (unlabeled samples with confidence scores higher than 0.9 are preserved, and indices corresponding to maximum confidence scores are treated as their pseudo-labels). Compared to RAMPL w/o MPLM^{-}, RAMPL w/o MPLM uses both labeled samples and unlabeled samples with pseudo-labels to construct a single class prototype for each category, which can effectively represent the complex data distribution of the HSI. The OAs of RAMPL w/o MPLM improve 10.05%, 8.10%, 3.68%, and 4.82% on the IP, H2013, SA, and KSC datasets, respectively. The proposed RAMPL carefully constructs multiple class prototypes for each category, which effectively model the intraclass diversity of different categories. Compared to RAMPL w/o MPLM* and RAMPL w/o MPLM, the OAs of our proposed RAMPL improve (24.19%, 13.28%), (16.19%, 8.82%), (8.72%, 4.93%), and (13.06%, 5.34%) on the IP, H2013, SA, and KSC datasets, respectively.

3) Impact of Proposed RACLM: To demonstrate the effectiveness of the proposed RACLM, ablation experiments are performed, and the results with five labeled samples in each category are illustrated in Table VII, where RAMPL w/o RACLM represents removing RACLM in the proposed RAMPL. Due to complex and diverse land-cover categories and the effect of external factors, the HSI exhibits complex spatial and spectral information, including spatial distribution differences and spectral variability. Existing pseudo-label learning-based methods fail to effectively extract discriminative features under a small number of labeled samples, reducing the reliability of generated pseudo-labels. The proposed RAMPL introduces RACLM to learn the spatial distribution difference and spectral variability of the HSI via a relation-aware pseudo-graph, improving the model discrimination and the reliability of generated pseudo-labels. Compared to RAMPL w/o RACLM, the proposed RAMPL achieves higher classification accuracy. In terms of OA, the proposed RAMPL improves 3.64%, 2.75%, 1.76%, and 1.18% on the IP, H2013, SA, and KSC datasets, respectively.

E. Parameter Analysis

1) Impact of Spatial and Spectral Masking Ratios: We employ spatial and spectral masking as enhancement strategies in RACLM and leverage spatial and spectral reconstruction losses to encourage RAMPL to mine local spatial and spectral correlations of neighboring pixels. We conduct extensive experiments to explore the impact of masking ratios in spatial and spectral dimensions on classification accuracy. Fig. 8 lists the results with five labeled samples in each category. In the spatial masking enhancement strategy, a larger spatial masking ratio means that more pixels within HSI patches will be masked, resulting in the model being unable to reconstruct the spatial–spectral features of these masked pixels through

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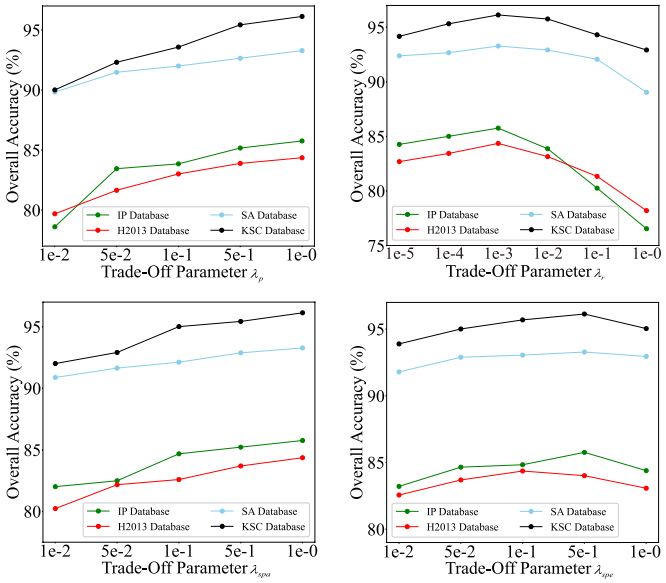


Fig. 9. Classification accuracy with different tradeoff parameters.

TABLE VIII
IMPACT OF THE NUMBER OF CLASS PROTOTYPES K ON CLASSIFICATION PERFORMANCE (OA: %)

Dataset	1	2	3	4	5	6	7
IP	72.48	82.16	85.76	84.96	80.03	71.61	68.53
H2013	75.54	81.99	84.36	86.69	87.93	89.43	90.44
SA	88.21	90.79	93.14	93.09	92.53	92.31	91.14
KSC	90.79	94.98	96.13	95.99	95.16	94.89	94.17

the remaining pixels. Conversely, a small spatial masking ratio may result in HSI patches containing more redundant information, which can also hinder the model from learning abundant features. According to the experimental results, the proposed RAMPL sets the spatial masking ratio to 10%, 10%, 60%, and 60% on the IP, H2013, SA, and KSC datasets, respectively. Similar to the spatial masking enhancement strategy, on four datasets, RAMPL achieves the best classification accuracy when the spectral masking ratio is set to 40%, 40%, 30%, and 30%, respectively.

2) Impact of Tradeoff Coefficients: On the IP, H2013, SA, and KSC datasets, we conduct extensive experiments to explore the impact of tradeoff coefficients, including λ_p, λ_r, λ_{spa}, and λ_{spe}, on classification performance. Fig. 9 records the OA with five labeled samples in each category. An appropriate set of tradeoff parameters can effectively train the weight parameters of the proposed RAMPL to improve classification accuracy. As shown in Fig. 9, when λ_p, λ_r, λ_{spa}, and λ_{spe} are set to 1.0, 0.001, 1.0, and 0.5 on the IP, SA, and KSC datasets, and when λ_p, λ_r, λ_{spa}, and λ_{spe} are set to 1.0, 0.001, 1.0, and 0.1 on the H2013 dataset, the proposed RAMPL achieves superior classification accuracy.

3) Impact of the Number of Class Prototypes K in Each Category: The proposed RAMPL constructs multiple class prototypes for each category to model intraclass diversity. The class prototypes are further used to generate high-quality pseudo-labels for optimizing RAMPL's weight parameters. To

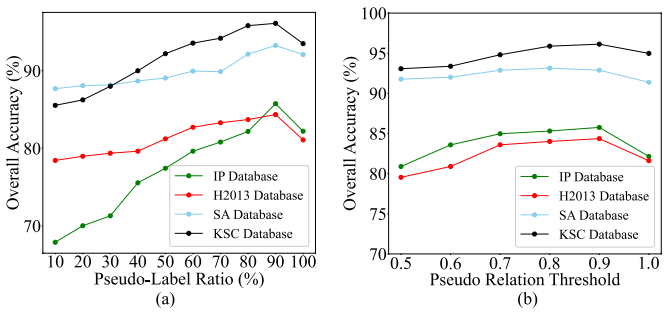


Fig. 10. Classification accuracy with different (a) pseudo-label ratios and (b) pseudo-relation thresholds.

explore the impact of the number of class prototypes K in each category on classification accuracy, we conduct extensive experiments, and the results with five labeled samples in each category are listed in Table VIII. An appropriate number of class prototypes facilitates modeling the intraclass diversity of different categories, thus improving classification accuracy. When too many class prototypes for each category are constructed, RAMPL is prone to overfitting the training data. Conversely, when the number of class prototypes is insufficient, RAMPL struggles to effectively represent intraclass diversity. According to the experimental results shown in Table VIII, when the number of class prototypes K in each category is set to 3 on different datasets, our RAMPL achieves promising classification performance.

4) Impact of Pseudo-Label Ratio γ: The proposed RAMPL generates high-confidence pseudo-labels for unlabeled samples whose distances between their spatial–spectral features and the class prototypes in the feature space are lower than the pseudo-label threshold τ, where τ is determined by the pseudo-label ratio γ in this article. These unlabeled samples and labeled samples are further used to optimize the model's parameters. We conduct extensive experiments to further explore the impact of the pseudo-label ratio γ on classification accuracy. Fig. 10(a) lists the results with five labeled samples in each category. A larger pseudo-label ratio γ indicates that the proposed RAMPL generates more pseudo-labels for unlabeled samples, which may contain more incorrect pseudo-labels in these samples. Incorrect pseudo-labels will confuse classification networks. Conversely, a smaller pseudo-label ratio γ will discard more high-confidence pseudo-labels, which can also lose more rich and diverse information. As shown in Fig. 10(a), the proposed RAMPL achieves the best classification accuracy when the pseudo-label ratio γ is set to 90% on four datasets.

5) Impact of Pseudo-Relation Threshold T: To further enhance feature discrimination, the RACLM is proposed to learn the spatial distribution difference and spectral variability of the HSI by decreasing the distance between samples with the same predictions in the feature space. A too-small pseudo-relation threshold T may cause samples from different categories to incorrectly learn similar spectral–spatial features, thereby confusing the classification model. Conversely, a too-large pseudo-relation threshold T results in our RAMPL failing to effectively explore the potential relations between

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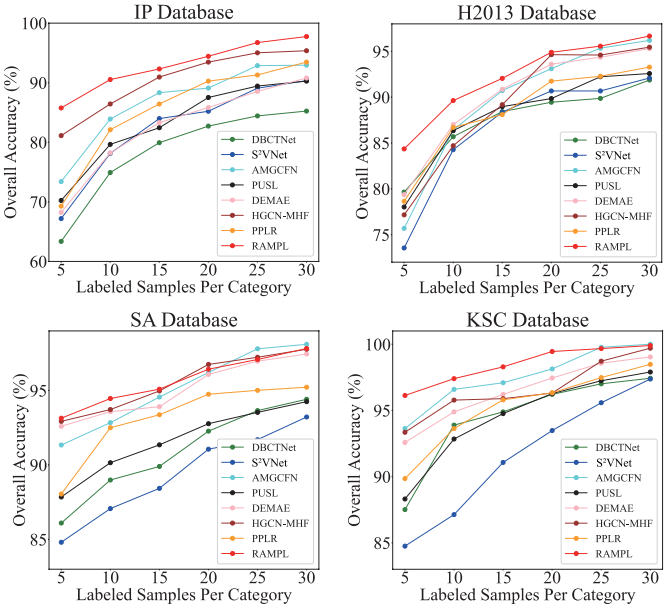


Fig. 11. Classification accuracy of the proposed RAMPL and the representative compared methods with different numbers of labeled samples.

samples, thereby limiting the classification performance. We conduct extensive experiments to explore the impact of pseudo-relation threshold T on the classification performance, and the results with five labeled samples in each category are illustrated in Fig. 10(b). We observe that the proposed RAMPL achieves the best classification accuracy when T is set to 0.9, 0.9, 0.8, and 0.9 on the IP, H2013, SA, and KSC datasets, respectively.

6) Impact of the Number of Labeled Samples in Each Category: We conduct extensive experiments to explore the classification accuracy of our proposed method and the representative compared methods under different labeled sample sizes, including {5, 10, 15, 20, 25, 30} labeled samples in each category. The experimental results on different datasets are illustrated in Fig. 11. As the number of labeled samples increases, different methods effectively capture diverse and discriminative spatial–spectral information of the HSI, thereby improving classification accuracy. As shown in Fig. 11, we can observe that as the number of labeled samples increases, the classification performance of the proposed method and other compared methods improves significantly, and the proposed method outperforms other methods under most labeled sample sizes. Moreover, these experimental results across different datasets strongly demonstrate the robustness and stability of the proposed method.

F. Complexity Analysis

To evaluate the computational complexity of the proposed RAMPL and the state-of-the-art compared methods, the training time, testing time, and parameter count are analyzed, and the results on the IP dataset are listed in Table IX. SSFTT, DBCTNet, and S^2VNet are supervised methods that only use a small number of labeled samples to optimize the model's weight parameters, thus consuming less training time

TABLE IX
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS ON THE IP DATASET

Method	Training Time (s)	Testing Time (s)	Parameters (M)
SSFTT	13.857	0.440	0.153
DBCTNet	57.849	0.781	0.030
S^2VNet	29.397	0.751	0.228
AMGCFN	18.562	0.225	0.368
SSLSM	464.901	2.211	36.779
PUSL	917.563	0.741	0.250
DSCA-Net	445.301	2.390	1.878
DEMAE	846.650	0.551	0.354
HGCN-MHF	14.559	1.643	0.169
RSEN	61.575	0.236	1.337
PPLR	743.449	0.462	0.250
RAMPL	1259.961	0.489	4.000

compared with other methods. AMGCFN and HGCN-MHF split the HSI into a series of irregular regions for feature extraction, thereby reducing the training cost compared with the proposed RAMPL. Since the remaining compared methods aim to leverage unlabeled samples to optimize weight parameters, they consume higher training time. The proposed RAMPL integrates MHSA and MSCB to extract both global and local features from different receptive fields and constructs multiple class prototypes for each category to generate high-quality pseudo-labels. As shown in Table IX, our RAMPL requires higher training time and parameters than other methods. In terms of testing time, the proposed RAMPL is lower than most compared methods and is competitive with SSFTT, AMGCFN, RSEN, and PPLR. Overall, our proposed RAMPL maintains competitive computational efficiency while achieving superior performance.

V. CONCLUSION

This article proposes an RAMPL method to enhance the reliability of generated pseudo-labels for accurately classifying the land-cover category of pixels. First, a TCM is designed to capture discriminative spatial–spectral features by exploiting both global and local information extracted by the attention mechanism and the proposed MSCB. Then, to generate high-quality pseudo-labels, we introduce an MPLM to model intraclass diversity by constructing multiple class prototypes for each category using unlabeled samples with pseudo-labels and a small number of labeled samples. Finally, we propose an RACLM to learn the spatial distribution difference and spectral variability of the HSI via a relation-aware pseudo-graph, thereby further enhancing spatial–spectral feature discrimination. Extensive experimental results on different datasets have verified the effectiveness of the proposed RAMPL.

In the future, we plan to investigate safety semi-supervised HSI classification methods that comprehensively consider the presence of unknown classes within unlabeled data. Due to the continuous change of land-cover categories and the high cost of manual labeling in remote sensing scenes, unlabeled data may fall into unknown classes that were not present in the labeled data. To accurately predict known classes and

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identify unknown classes, efficiently leveraging unlabeled data to enhance feature discrimination is a challenging problem.

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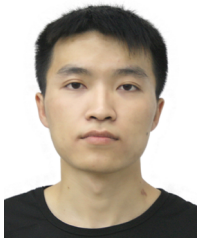
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